Structural, Single-Cell and Spatial Analyses with an AI Agent Workspace: EGFR Variants, PBMC3k Robustness and Mouse-Brain Spatial Domains

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Abstract

We used an AI agent workspace (Claude Code, model Claude Opus 5.5) to carry out three analyses and audited its work. **(1)** For EGFR variants G719S, T790M and L858R we mapped canonical positions onto PDB 4HJO (whose numbering is canonical −24) and measured their structural context. L858R has the most direct structural support for a mechanism (burial in the inactive state), T790M sits at the drug pocket but a drug-bound Met790 structure shows no steric clash, and G719S evidence is indirect. **(2)** PBMC3k clustering gives nine populations. B cells are robust to resolution and subsampling; the CD8 T/NK split depends on resolution; the naive/memory CD4 boundary is a continuum, and the "naive CD4" cluster contains an estimated 31–40% CD8 T cells. **(3)** Expression-only Visium domains are strongly spatially coherent (same-domain edge fraction 0.783 vs 0.084 under 1,000 label permutations, $p = 0.001$), and this survives depth/mitochondrial regression (ARI 0.78). Moran's I and Wilcoxon DE rankings agree only moderately ($\rho = 0.52$), and less when DE is ranked by fold change ($\rho = 0.29$). All results are regenerated by one script in about four minutes on a CPU.

1 Introduction

The assignment asks whether an agent workspace can produce biomedical analyses that are correct, checkable and honestly qualified. We address three tasks: a protein-structure interpretation of three clinically important EGFR mutations, a robustness study of single-cell clustering, and a test of whether expression-defined spatial domains are organised in tissue. Throughout, claims are tagged as observations computed here [OBS], predictions or interpretations [PRED], or retrieved external evidence [EXT]. The agent's own errors, and the checks that caught them, are reported in Appendix B.

2 Methods

Workspace and reproducibility. The agent wrote and ran all code (`submission/*/code/`) on a CPU-only Linux machine; no GPU or structure prediction was run. `run_all.sh` creates a Python 3.12 environment with `uv` from `requirements.txt`, downloads and SHA-256-checks every input (`fetch_data.sh`), runs the 11 analysis scripts in order and compiles this report. Key software: scanpy 1.12.5, squidpy 1.8.4, leidenalg 0.12.0, scikit-learn 1.9.1, gemmi 0.7.5, Biopython 1.88, freesasa 2.2.1, PyMOL 3.2.

Protein (Task 1). UniProt P00533 (entry v301) supplied the canonical sequence and annotations. Each structure's full SEQRES was aligned globally to UniProt (BLOSUM62) to map residues; the depositors' SEQADV records were used as a cross-check. Structures: 4HJO (primary; WT+V948R, erlotinib, inactive, 2.75 Å [11]), 1M17 (WT, erlotinib, active-like [13]), 2ITX/2ITN/2ITV (WT/G719S/L858R with AMP-PNP [17]), 2JIT (T790M apo [18]), 4I22 (L858R/T790M/V948R, gefitinib [2]) and the AlphaFold DB model [4]. Measurements use heavy atoms only: minimum inter-atomic distance, residues within 5 Å, relative SASA (freesasa, 1.4 Å probe, ligands removed) and αC state from the K745 NZ–E762 OE distance (< 4 Å = αC-in).

Single cell (Task 2). PBMC3k (10x Genomics [20]). QC: 200–2,500 genes, $< 5\%$ mitochondrial reads, genes in ≥ 3 cells. Then normalisation to 10^4, log1p, 2,000 HVGs (Seurat flavour), regression of total counts and % mito, scaling, PCA (30 PCs), kNN ($k = 15$), Leiden [14] and UMAP, all with seed 0 [16]. Labels come from Wilcoxon markers computed on these data plus a rule that scores literature marker sets per cluster and reports a cluster as ambiguous if the top two scores differ by < 0.15; no reference labels were used. Robustness: Leiden at resolutions 0.3, 0.6, 1.0, 1.4 on the same graph; a 70% cell subsample (seed 42) re-run from raw counts; extra Leiden seeds and subsample seeds 1–5.

Spatial (Task 3). Visium V1 adult mouse brain (10x Genomics). QC: $\geq$ 5,000 UMIs, $\geq$ 1,000 genes, $< 30\%$ mito. Domains were found by clustering an AnnData object built from counts only (assertions check that no coordinates or image are present): seurat_v3 HVGs, PCA (30), kNN ($k = 15$), Leiden (resolution 0.5, seed 0). The physical-neighbour graph is the Visium hexagonal lattice (squidpy [9] `spatial_neighbors`, grid, 6 neighbours, first ring). The main statistic is the fraction of graph edges joining two spots of the same domain. The null permutes domain labels across spots on the fixed graph (1,000 times, seed 0; one-sided empirical $p = (1 + \#\{\text{null} \geq \text{obs}\})/(N + 1)$). Spatially variable genes: Moran's I on the 200 top HVGs; DE: Wilcoxon one-vs-rest across domains on the same genes.

3 Results

3.1 EGFR: G719S, T790M and L858R

Mapping [OBS]**.** All three wild-type residues match UniProt. 4HJO and 1M17 use mature-protein numbering (canonical -24; L858 is Leu834 in 4HJO), while the 2IT*/2JIT/4I22 entries use canonical numbering. In every structure the only mismatches against UniProt are the deposited mutations, including the engineered V948R of 4HJO, which lies 26–31 Å from all three sites (Appendix Table 5). AlphaFold confidence is high at T790 (pLDDT 88) but low at L858 (51), consistent with a conformationally variable activation loop.

Structural context [OBS]**.** In inactive 4HJO, **T790** is the residue closest to erlotinib (3.48 Å) and is the closest of the three sites to the ligand in all six ligand complexes. **L858** is buried (relative SASA 0.10 vs 0.15–0.40 in αC-in structures), 2.96 Å from DFG-D855 and 4.57 Å from αC residue M766, in a mostly hydrophobic shell that also contains K745, D855 and K860. **G719** is the first Gly of the P-loop, solvent-exposed (SASA 0.35) and 5.06 Å from erlotinib. Variant-specific checks: superposing apo Met790 (2JIT) onto 4HJO suggests a clash with erlotinib (0.6–1.3 Å), but drug-bound Met790 in 4I22 sits 3.3–3.5 Å from the drug, so the clash is a property of the apo rotamer; in the active-like 1M17 pocket the 4I22 side chain comes to 2.5 Å. In G719S (2ITN), Ser OG is 5.15 Å

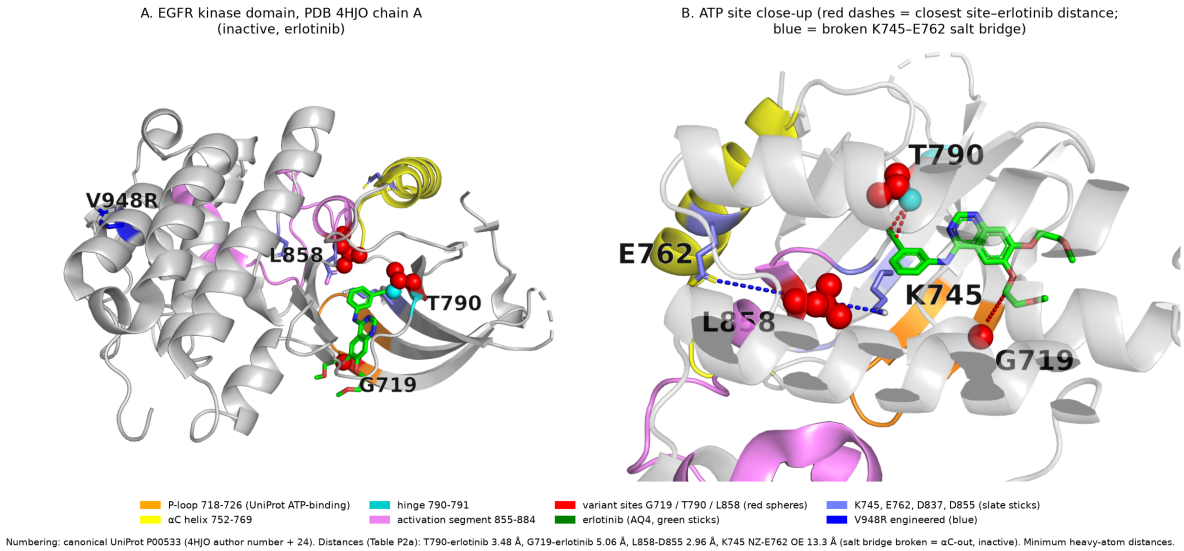


Figure 1: Annotated EGFR kinase domain (PDB 4HJO, canonical numbering). (A) Overview; (B) ATP site. Red spheres: variant sites. Green: erlotinib. Red dashes: closest site–erlotinib distances (T790 3.48 Å, G719 5.06 Å). Blue dashes: broken K745–E762 salt bridge (13.3 Å, αC-out). Interactive version: `egfr_4hjo_interactive.html`.

from the nearest phosphate oxygen (no direct hydrogen bond), and the wild-type Gly backbone angles are not glycine-specific. 4I22 is itself αC-out yet accommodates Arg858 through salt bridges to D855 and D837, which argues against a simple "Arg858 is incompatible with the inactive state" model.

Interpretation. Table 1 ranks the variants by how directly the structures support a mechanism: **L858R > T790M > G719S**. Ranking by functional impact on one scale is not meaningful, because L858R and G719S are activating mutations that predict inhibitor sensitivity [7, 8], while T790M confers resistance. AlphaMissense [1] scores all three as pathogenic (≥ 0.96) and cannot separate these classes.

3.2 PBMC3k: cell types and their robustness

Baseline [OBS]**.** QC kept 2,638 of 2,700 cells (13,656 genes). At resolution 1.0, Leiden gives nine clusters (Fig. 2A, Table 2). Resolution 1.0 is the lowest tested value that separates CD8 T from NK cells; it was chosen after seeing results at 0.6, which we disclose as a researcher degree of freedom.

Resolution and subsampling [OBS]**.** Table 3 summarises both checks. Across resolutions, *B cells* are perfectly stable (recall and purity 1.00). NK and CD14 monocyte cells always keep their label, but at low resolution their clusters absorb CD8 T cells (NK purity 0.36–0.38) or FCGR3A monocytes and DCs. CD8 T, naive CD4 and memory CD4 exist only at resolution ≥ 1.0. Changing only the Leiden seed gives ARI 0.77–0.83, mostly from the 2-iteration default (ARI 0.94 when run to convergence); nearly all seed-to-seed change is at the naive/memory boundary. In the 70% subsample (ARI 0.65, label agreement 77.6%), CD8 T cells split between NK (55%) and memory CD4 (43%), and the 8 retained platelets joined the FCGR3A monocytes. Re-clustering the same subsample at 1.2, or using five other subsample seeds, recovers CD8 T every time; platelets return in only 1 of 5.

Table 1: Comparison of the three EGFR variants (measurements from 4HJO unless stated; `table_p2a`, `table_p2d`, `table_p2e`).

Variant	Structural evidence [OBS]	Proposed mechanism [PRED] / support [EXT]	Class	Confidence
L858R	A-loop helix, buried (SASA 0.10 inactive vs 0.15–0.40 active); 2.96 Å to D855, 4.57 Å to αC M766; pLDDT 51	Arg in the pocket that packs the inactive A-loop against αC shifts the balance toward the active state [17, 12]. 4I22 shows Arg858 can be accommodated in an αC-out state, so it is a shift, not an incompatibility	Activating	Mod.–high
T790M	Gatekeeper; 3.48 Å to erlotinib; apo Met clashes (0.6–1.3 Å) but drug-bound Met does not (3.3–3.5 Å)	Resistance mainly via increased ATP affinity [18, 5, 10]; our data do not support a purely steric block	Resistance	Moderate
G719S	P-loop Gly (GxGxxG), exposed (SASA 0.35); 5.06 Å to erlotinib; Ser OG 5.15 Å to phosphate; P-loop shift $\leq$1.95 Å	Altered P-loop flexibility destabilises the inactive state [17]; no new contact or backbone strain observed	Activating	Low–mod.

Ambiguous population: naive vs memory CD4 [OBS] [PRED]**.** Clusters 0 and 1 differ only modestly (cluster 1: ribosomal genes, $\log_2$FC 0.25–0.5; cluster 0: S100A4 −1.7, S100A11 −1.5, IL32 −1.0). The naive-minus-memory score is not bimodal (bimodality coefficient 0.39 $<$ 0.555; AUC between clusters 0.73), and 16–21% of each cluster's kNN edge weight points to the other cluster (Fig. 2C). This matches the continuous naive → central memory → effector memory differentiation of CD4 T cells. Cluster 1 also contains CD8 cells: 26% express CD8A or CD8B, against a background of 3–10% in B and NK cells and only 60% detection in true CD8 T cells, giving an estimated 31–40% CD8 T cells. Naive CD8 cells share the CCR7/SELL/LEF1 programme and lack cytotoxic genes, so they co-cluster with naive CD4 cells; CD4 mRNA itself is detected in only 9–12% of CD4 T cells. We therefore relabel cluster 1 "Naive T, CD4/CD8 mixed". Protein markers (CITE-seq) would be needed to resolve it.

3.3 Visium mouse brain: spatial domains

Domains [OBS]**.** QC kept 2,683 of 2,702 in-tissue spots (17,535 genes). Expression-only clustering gave 16 domains (20–339 spots) that map to contiguous regions (Fig. 3A). Tentative anatomy from marker genes and the Allen atlas [6] [PRED]: cortical layers (domains 2, 3, 15), thalamus (4, 5; Tcf7l2, Prkcd), fibre tracts (8; Mbp, Plp1), hippocampus (11–13), hypothalamus (6), ventricle/ependyma (14), meninges (9) and striatum-like (1, probably also containing choroid plexus). Domains 0, 7 and 10 are not annotated with confidence.

Spatial organisation and negative control [OBS]**.** The hexagonal graph has 7,759 edges (mean degree 5.78). 78.3% of edges join spots of the same domain, against 8.4% under label permutation (z = 224; no permutation reached the observed value, p = 0.001, the minimum for N = 1,000; Fig. 3B, Table 4). Every domain is 10–98× more self-adjacent than expected, and each domain's indicator has Moran's I 0.34–0.88. The strongest between-domain contacts are between hippocampal sub-fields. Sequencing depth and % mito are themselves spatially structured (Moran's I 0.49

Table 2: Baseline clusters and labels (resolution 1.0; markers are Wilcoxon $\log_2$FC from these data, `table_s2b`; marker biology from [20, 3, 15]).

Cluster	Cells	Top data-derived markers	Label
0	508	IL32, LTB, CD3D, IL7R, S100A4	Memory CD4 T
1	643	ribosomal genes, LDHB, CD3D; CCR7 in 44%	Naive T, CD4/CD8 mixed (revised, §3.2)
2	342	CD79A, MS4A1, CD79B	B
3	157	FCGR3A, LST1, AIF1	FCGR3A$^+$ monocyte
4	155	GNLY, GZMB, NKG7, PRF1	NK
5	306	CCL5, NKG7, GZMK, CD3D; CD8A in 51%	CD8 T
6	475	S100A8, S100A9, LYZ, FCN1	CD14$^+$ monocyte
7	38	FCER1A, CST3, HLA class II	Dendritic cell
8	14	PPBP, PF4	Platelet

Table 3: Robustness checks (`table_s3b/c/e/f`, `table_s4a/b/d/e`).

Comparison	Clusters	ARI	NMI	Main change
Resolution 0.3 vs 1.0	5	0.59	0.77	T, NK/CD8 and myeloid clusters merge
Resolution 0.6 vs 1.0	7	0.64	0.82	CD8 T merges with NK; naive+memory CD4 merge
Resolution 1.4 vs 1.0	11	0.70	0.80	naive/memory CD4 re-split (72% / 82% kept)
Leiden seeds 1–5	9	0.77–0.83	–	naive/memory boundary (0.70–0.93 kept)
70% subsample (seed 42)	7	0.65	0.75	CD8 T and platelets lost; agreement 77.6%
Subsample seeds 1–5	8–9	–	–	CD8 T 5/5, platelets 1/5; agreement 81–85%

and 0.58) and partly explained by domain (η^2 36% and 31%). Regressing both out and re-clustering gives 13 domains that agree with the originals (ARI 0.78) and are equally coherent (0.792 vs 0.093).

Spatial genes vs differential expression [OBS] [PRED]**.** All 200 tested HVGs have FDR < 0.05 for Moran's I, which is expected for pre-selected variable genes, so we use ranks rather than the count. The top spatial genes are Prkcd ($I = 0.87$, thalamus), Mbp (fibre tracts), Ttr (choroid plexus) and Pmch (Appendix Fig. 4); blood and vascular genes (Ccl5, S100a8/9) rank lowest. Genes high in both rankings mark large regions that match a domain (Prkcd, Mbp). *Agrp* is spatially sharp (Moran rank 7) but ranks 183rd by Wilcoxon z: it is confined to the small arcuate region, so most spots of its best domain are zero and the rank-based test penalises it, even though its fold change is large ($\log_2$FC 5.5, rank 27). Conversely, Cpne9 separates domains on average (DE rank 30) but varies in a graded way inside them (Moran rank 103). The rankings therefore measure different things, local neighbour agreement vs separation of predefined groups, and their agreement depends on the DE metric.

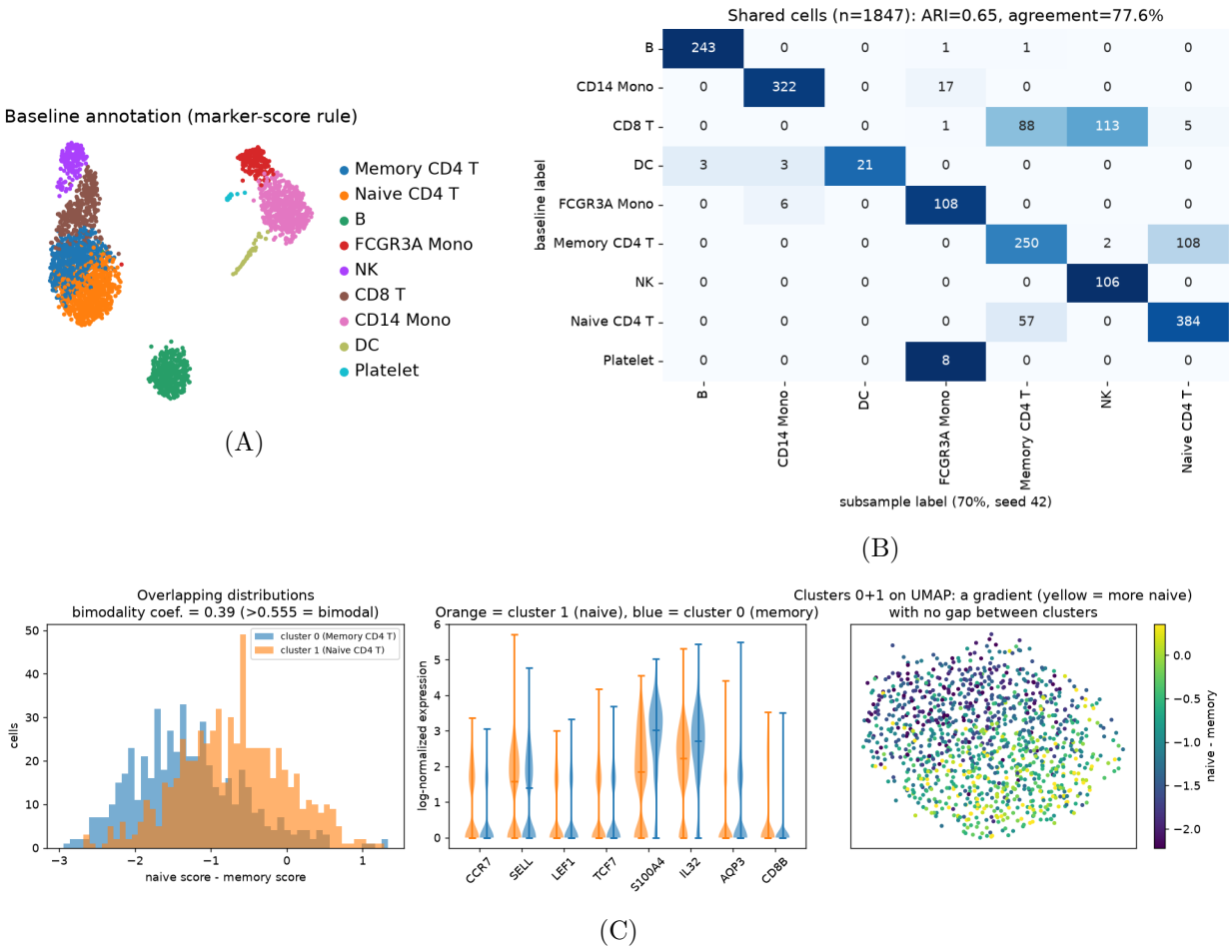
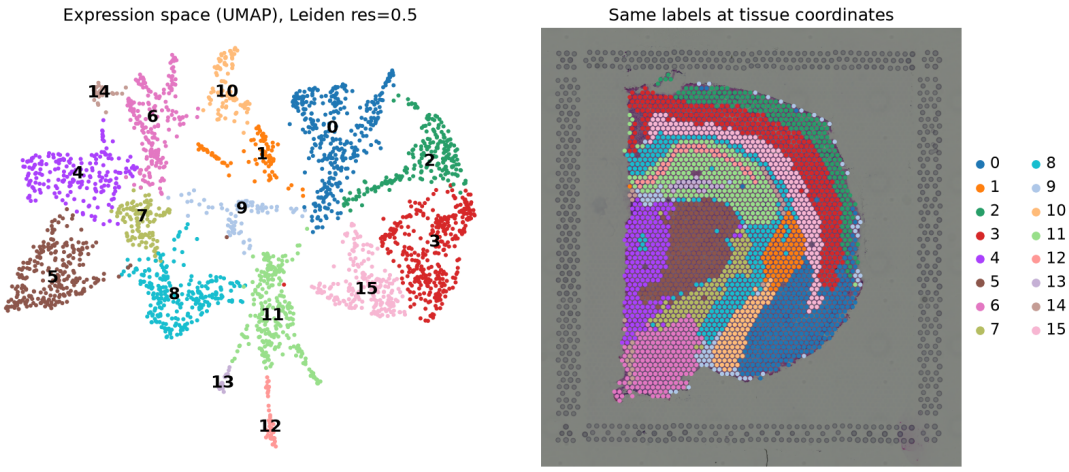


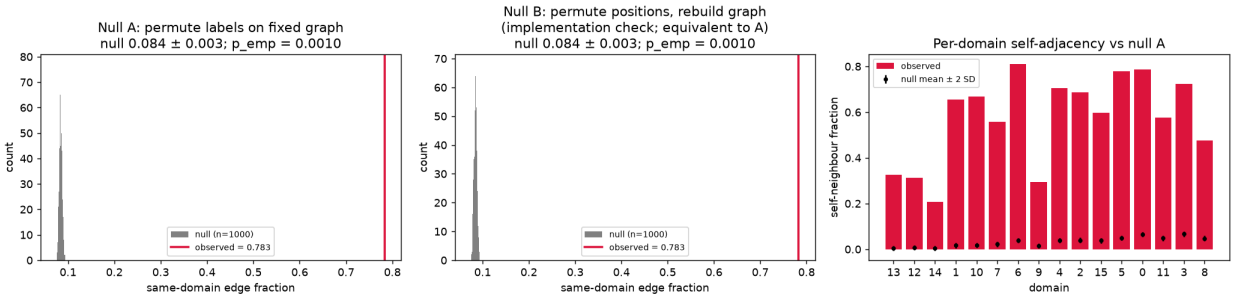
Figure 2: PBMC3k. (A) Baseline UMAP with automatic labels. (B) Baseline vs 70% subsample labels on the 1,847 shared cells. (C) Naive-minus-memory score for clusters 0 and 1: overlapping distributions, graded markers and a gradient on the UMAP without a gap.

Table 4: Spatial statistics (`table_t5a`, `table_t5c`, `table_t4b`).

Statistic	Observed	Null mean ± SD	z	Empirical p
Same-domain edge fraction (16 domains)	0.783	0.084 ± 0.003	224	0.001
Same, after depth/mito regression (13 domains)	0.792	0.093 ± 0.003	–	0.001
Spearman ρ: Moran's I vs DE (Wilcoxon z)	0.52			1.5×10^{-15}
Spearman ρ: Moran's I vs DE (max $\log_2$FC)	0.29			$< 10^{-4}$



(A)



(B)

Figure 3: Visium domains. (A) The same expression-only labels in UMAP space and on tissue. (B) Observed same-domain edge fraction (red) vs 1,000 label permutations; the middle panel is the graph-rebuild implementation check, equivalent to the left null; right: per-domain self-adjacency vs null mean ±2 SD.

4 Discussion and limitations

Across tasks the most reliable conclusions are the coarse ones: the location of the three EGFR variants relative to the drug and the catalytic machinery, the major PBMC lineages, and the existence of spatially coherent brain domains. Finer conclusions depend on choices that we varied explicitly: the L858R mechanism rests on static structures that carry an engineered V948R, the CD8 T/NK split and the naive/memory boundary depend on resolution and sampling, and the gene-ranking comparison depends on the DE metric. The main limitations are: (i) no single-mutant T790M–erlotinib complex was analysed and mechanisms are inferred from static structures; (ii) PBMC labels rest on sparse mRNA, so CD4/CD8 identity cannot be resolved without protein data; (iii) the spatial null assumes complete spatial randomness, a weak alternative (a null preserving autocorrelation, such as shifted label maps, was not run), Visium spots mix several cells, and only one section was analysed.

Code and data availability. Code, outputs and this report: `https://github.com/CosmoBean/GenAIBiomedicine`. Run `bash submission/run_all.sh`. Seeds: Task 2 seed 0 (PCA, kNN, Leiden, UMAP), subsample seed 42, extra seeds 1–5; Task 3 seed 0 (clustering, null), seed 1 (graph-rebuild check); Task 1 is deterministic. Data: 10x Genomics PBMC3k and V1_Adult_Mouse_Brain (course repository copies, checksummed); UniProt P00533, P04626, P21860, Q15303; PDB 4HJO, 1M17, 2ITX, 2ITN, 2ITV, 2JIT, 4I22; AlphaFold DB AF-P00533-F1 v6.

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A Checkpoint evidence index

ID	Status	Evidence (page; figure/table; code)
P1	complete	p. 2 (Mapping), Table 5; `protein/code/01_map_and_measure.py`, `table_p1a--c`
P2	complete	p. 4, Tables 1, 6; `01_map_and_measure.py`, `02_mechanism_checks.py`, `table_p2a--f`
P3	complete	Table 1 and its interpretation (p. 4); references
P4	complete	Fig. 1 (p. 3), Table 1; `03_render_structure.py`
S1	complete	Methods; Table 2, Fig. 2A (p. 6); Fig. 5; `single_cell/code/01_qc_baseline.py`
S2	complete	Table 2 (p. 5); `table_s2a--d`, `dotplot__s2c_canonical_markers.png`
S3	complete	Table 3 (p. 5); `02_resolution_sweep.py`, `table_s3a--f`, `fig_s3a`
S4	complete	Table 3, Fig. 2B; `03_downsample.py`, `table_s4a--e`
S5	complete	§3.2, Fig. 2C, Table 7; `04_ambiguous_population.py`, `table_s5a--d`
T1	complete	Fig. 3A (p. 7); `spatial/code/01_expression_domains.py`, `table_t1*`
T2	complete	Fig. 3B, Table 4 (p. 6); `02_spatial_statistics_and_null.py`, `table_t2a--d`
T3	complete	Fig. 4; `03_svg_vs_de.py`, `table_t3a--b`
T4	complete	Table 4, Table 8, Fig. 4; `table_t3b_t4_moran_and_DE_rankings.csv`, `table_t4b`
T5	complete	Fig. 3B, Table 4; `02_spatial_statistics_and_null.py`, `04_depth_confound_check.py`, `table_t5a--d`
A	complete*	Appendix B (*student verification boxes to be completed by the student)
R	complete	`README.md`, `run_all.sh`, `fetch_data.sh`, `requirements.txt`; seeds in §4

B Agent audits

Agent workspace: Claude Code (Anthropic) with model Claude Opus 5.5, with shell, Python and web access. After the analysis, three independent reviewer agents (one per task) re-derived the key numbers from scratch; issues they found are listed as weaknesses below.

B.1 Task 1 (EGFR)

Sources. UniProt P00533 (v301) and paralogs P04626, P21860, Q15303; RCSB PDB 4HJO, 1M17, 2ITX, 2ITN, 2ITV, 2JIT, 4I22; AlphaFold DB AF-P00533-F1 v6 (model, pLDDT, AlphaMissense); 13 papers, each verified on PubMed.

Computed vs retrieved. *Computed:* sequence checks, SEQRES-to-UniProt mapping, all distances, neighbour shells, SASA, B-factors, backbone angles, αC state, superpositions, paralog alignment. *Retrieved:* sequences and annotations, coordinates, SEQADV records, pLDDT, AlphaMissense scores, and the published mechanisms.

Independent checks. (1) Mapping vs the depositors' SEQADV records: 4HJO lists Arg924 as engineered V948, confirming the -24 offset, and 2ITN/2ITV/2JIT list S719/R858/M790 at canonical numbers; all agree. (2) All 13 citations resolved to the expected title and journal on PubMed. One newly added PMID was wrong and was corrected by this check.

Weaknesses found. *Mapping bug:* aligning only resolved residues gave an inconsistent offset for 2ITX/2ITV; fixed by aligning full SEQRES. *Misleading first result:* apo Met790 superposition suggested a steric clash; tested against drug-bound 4I22, which shows none. *Selective description (review):* L858's charged neighbours and the 4I22 counter-evidence were initially omitted; both are now reported and confidence lowered. *Cut-offs:* M766 at 4.57 Å sits near the 4.5/5 Å contact thresholds.

Student verification (to complete before submission): open 4HJO at RCSB, confirm Leu834 is canonical L858, measure Thr766–erlotinib in Mol* and compare with 3.48 Å. Result: _____

B.2 Task 2 (PBMC3k)

Sources. PBMC3k filtered matrix (10x Genomics via course repository, SHA-256 checked); marker biology from [20, 3, 15]; methods [16, 14]. No reference labels or cell-type databases.
Computed vs retrieved. *Computed:* QC, normalisation, HVGs, PCA, kNN, UMAP, Leiden, Wilcoxon markers, marker-set scores, ARI/NMI, contingency tables, subsampling, seed sweeps, bimodality coefficient, kNN mixing, CD8 detection rates. *Retrieved:* canonical marker lists and the biology linking markers to cell types.
Independent checks. (1) Seed-only control: ARI 0.77–0.83 between seeds (0.94 when Leiden runs to convergence), which calibrates the resolution and subsample effects. (2) The CD8 loss after subsampling was re-tested at resolution 1.2 and with five subsample seeds: CD8 T was recovered every time, so it is a threshold effect, not a missing population.
Weaknesses found. *Mislabelled cluster:* the rule called cluster 1 "Naive CD4 T"; 26% of its cells express CD8 (31–40% after detection correction), so it was relabelled. The first fix ("CD4-dominant") ignored CD8 dropout and was corrected after review. *Post-hoc resolution:* 1.0 was chosen after seeing 0.6. *Overclaimed stability (review):* recall-only stability hid NK clusters that are 62–64% CD8 T at low resolution; purity was added. *Figure errors:* Fig. 2C first had colours reversed between panels and a saturated colour scale; both fixed.

Student verification (to complete before submission): from `table_s2b_top10_markers.csv` alone, assign clusters 2, 4 and 6; then rerun `03_downsample.py` and confirm the 77.6% agreement. Result: _____

B.3 Task 3 (Visium)

Sources. 10x Genomics V1 Adult Mouse Brain matrix and spatial archive (course repository, SHA-256 checked); squidpy [9]; Allen Mouse Brain Atlas [6].
Computed vs retrieved. *Computed:* QC, HVGs, PCA, kNN, Leiden domains, UMAP, hexagonal graph, same-domain edge fraction, neighbourhood enrichment, Moran's I, Wilcoxon DE, rank correlations, permutation null, depth/mito regression. *Retrieved:* anatomical meaning of marker genes.
Independent checks. (1) The permutation mean (0.0844) matches the analytic expectation $\sum_k n_k(n_k-1)/(N(N-1)) = 0.0844$, confirming the null code. (2) A reviewer rebuilt the hexagonal graph by hand from array coordinates and obtained the same 7,759 edges and statistic 0.7832. (3) The absence of coordinates during clustering is enforced by assertions.
Weaknesses found. *Null B bug (review):* the coordinate-shuffle null rebuilt its graph from array coordinates, on which the grid mode keeps only 4 neighbours (5,178 edges), so it was compared on a different graph (z 182 instead of 224). It now uses the same coordinates, asserts an identical edge count, and is reported as an implementation check because it is equivalent to label permutation. *DE-metric dependence (review):* only the Wilcoxon-z ranking was reported; the fold-change ranking lowers ρ from 0.52 to 0.29 and is now shown. *Weak null:* complete spatial randomness only. *Anatomy:* inferred from markers, not checked against atlas images.

Student verification (to complete before submission): compare two domains in Fig. 3A with the Allen Brain Atlas coronal reference and their markers in `table_t1c`; record whether the labels hold. Result: _____

C Supplementary tables and figures

Table 5: Structure mapping (`table_p1b`). Offset = author − canonical numbering.

PDB	Content	Offset	G719 / T790 / L858 residue	Mismatches vs UniProt
4HJO	WT(+V948R), erlotinib, inactive	−24	Gly695 / Thr766 / Leu834	R948 (engineered)
1M17	WT, erlotinib, active-like	−24	Gly695 / Thr766 / Leu834	none
2ITX	WT, AMP-PNP	0	Gly / Thr / Leu	none
2ITN	G719S, AMP-PNP	0	**Ser** / Thr / Leu	S719
2ITV	L858R, AMP-PNP	0	Gly / Thr / **Arg**	R858
2JIT	T790M, apo	0	Gly / **Met** / Leu	M790
4I22	L858R/T790M/V948R, gefitinib	0	Gly / **Met** / **Arg**	M790, R858, R948
AF v6	AlphaFold model; pLDDT	0	73.3 / 88.4 / 51.2	none

Table 6: Site measurements in three reference structures (Å; full table for 8 structures in `table_p2a`).

Structure	Site	→ ligand	→ K745 NZ	→ E762	→ D855	→ αC	rel. SASA
4HJO (K–E 13.3)	G719	5.06	7.78	20.35	10.02	16.5	0.35
	T790	3.48	9.03	14.27	8.57	7.5	0.21
	L858	5.26	3.58	6.18	2.96	4.57	0.10
1M17 (K–E 3.05)	G719	4.50	9.57	12.41	10.89	12.3	0.37
	T790	3.41	8.56	7.38	9.03	3.2	0.17
	L858	10.12	8.70	6.64	5.04	6.6	0.15
2ITX (K–E 3.58)	G719	4.27	10.17	13.10	11.93	13.0	0.35
	T790	3.13	7.10	6.62	9.85	3.9	0.15
	L858	6.27	10.17	6.81	4.94	6.2	0.25

Table 7: Marker detection (% of cells) in T-cell clusters (`table_s5b`, `table_s5d`).

Genes	Cluster 0 “memory”	Cluster 1 “naive”	Cluster 5 CD8 T
CCR7 / SELL / LEF1	27 / 51 / 28	44 / 60 / 35	6 / 31 / 7
S100A4 / IL32 / AQP3	91 / 91 / 38	69 / 78 / 26	96 / 96 / 19
CD4	12	9	4
CD8A or CD8B	11	26	60
GZMK / CCL5	9 / 28	5 / 18	57 / 97

Table 8: Spatial vs DE ranking examples (rank among 200 genes; `table_t3b_t4_moran_and_DE_rankings.csv`).

Category	Gene	Moran rank	DE rank (z)	DE rank ($\log_2$FC)	% spots detected
High in both	Prkcd	1	1	41	53
	Mbp	2	2	102	100
Spatial-high, DE(z)-low	Agrp	7	183	27	4
	Gm5741	10	190	97	4
	Fezf1	22	161	62	5
DE-high, spatial-lower	Cpne9	103	30	171	42

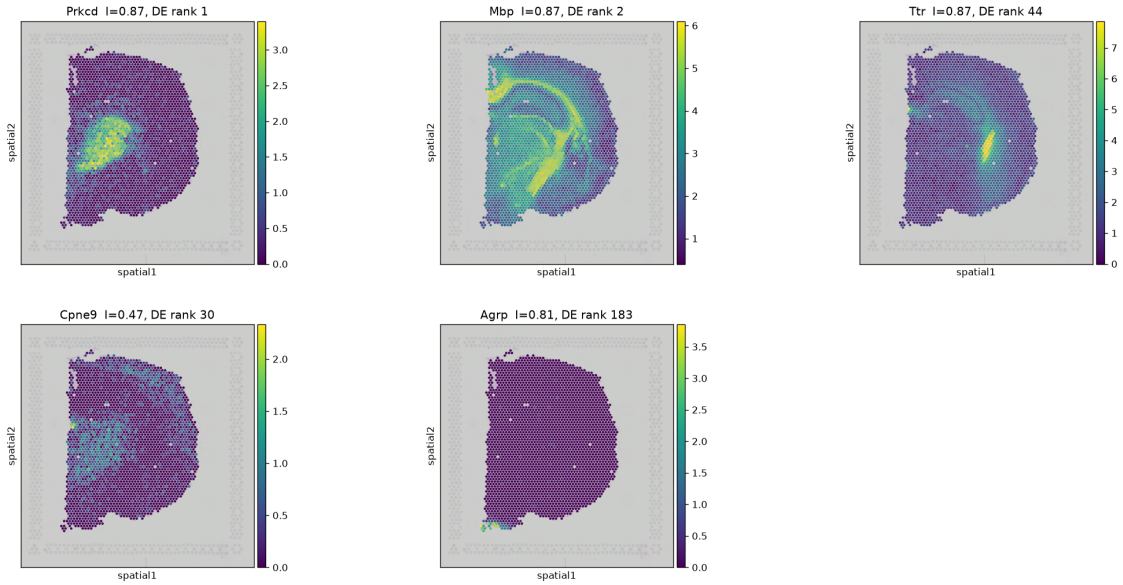


Figure 4: Selected genes on tissue, with Moran's I and Wilcoxon-z DE rank.

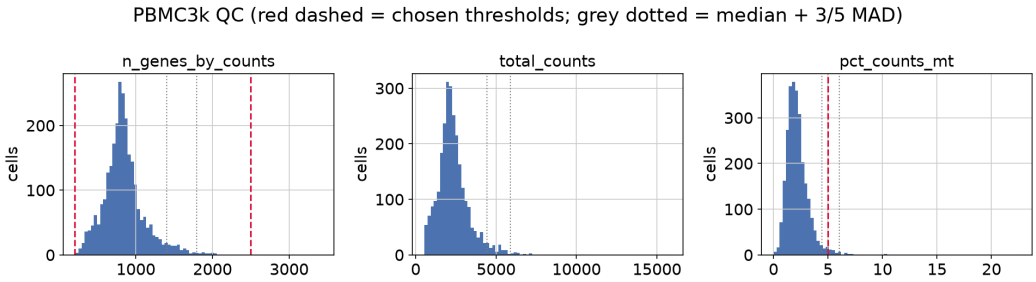


Figure 5: PBMC3k QC distributions with thresholds (red) and median + 3/5 MAD (grey). Medians: 817 genes, 2,197 UMIs, 2.03% mito; the cuts remove 5 cells (>2,500 genes) and 57 cells (≥5% mito).